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DAY : Sunday

TIME :   DATE : 28/06 /2026

PHY 209

Mid → 14/07

Quiz → 2 → 07/07

Ques: ✦ 1D কিনা থাকবে লেখো. (1D)

Time dependent Schrödinger Equation :

1D ← (1 Dimentional as only (x))

ψ depends on time (t) & position (x')

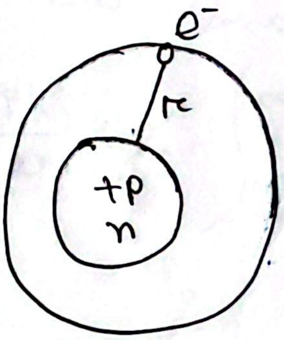
Q. Write down the 1D Schrödinger eqn] — 1

Wave function

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V\psi(x,t) \quad \text{— (i)}$$

V ↳ potential

$\hbar = \frac{h}{2\pi}$



Potential usally function of position.

$$V = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \quad e^2 \rightarrow e^- \text{ চার্জ}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

It is a partial differential equation.

Separation of variable.

Let, $\psi(x,t) = \psi(x)\,\phi(t) \rightarrow \frac{\partial \psi}{\partial t} = \psi(x) \frac{\partial \phi}{\partial t}$

*[Variable Separation]

~~$f(x,y)$~~

$f(x,y) = \underbrace{2x^2}_{g(x)}\ \underbrace{y}_{P(y)}$

$f(x,y) = g(x)P(y)$

$$\frac{\partial \psi(x,t)}{\partial t} = \phi(t) \frac{\partial \psi}{\partial x} \quad \frac{\partial \psi}{\partial x}$$

$$\frac{\partial^2 \psi}{\partial t} = \phi(t) \frac{\partial^2 \psi}{\partial x^2}$$

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From eqn (i) we get,

(ii) Left side is t fun & Right side is x fun

~~$i\hbar\,\phi(t)\frac{\partial\psi}{\partial x} = -\frac{\hbar^2}{}$~~

$$i\hbar\,\psi(x)\frac{\partial\phi}{\partial t} = -\frac{\hbar^2}{2m}\phi(t)\frac{\partial^2\psi}{\partial x^2} + V\psi(x)\phi(t)$$

Divide by $\psi\phi$ on both side,

$$\underbrace{i\hbar\frac{1}{\phi}\frac{\partial\phi}{\partial t}}_{\text{function of time only}} = \underbrace{-\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{\partial^2\psi}{\partial x^2} + V}_{\text{function of position only}}$$

They must be equal to some constant

$$\frac{i\hbar}{\phi}\frac{\partial\phi}{\partial t} = E \quad (\text{constant})$$

$$-\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{\partial^2\psi}{\partial x} + V = E \quad (\text{constant})$$

$$\Rightarrow i\hbar\frac{d\phi}{dt}\,E\phi \quad \text{——— (ii)} \longrightarrow \text{fun of time only}$$

$$\Rightarrow \boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi} \quad \text{——— (iii)} \longrightarrow \text{Time independent Schrödinger eq}^n$$

Q. Write down the dependend eqn and Time independent eq^n

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eqn (ii) ⇒

$$i\hbar \frac{d\phi}{dt} = E\phi$$

or, $i\hbar \frac{d\phi}{\phi} = E\,dt$

or, $i\hbar \int \frac{d\phi}{\phi} = \int E\,dt$

or, $i\hbar \ln\phi = Et + c$

$* \quad \frac{dy}{dx} = 2$

$\Rightarrow dy = 2\,dx$

$\Rightarrow \int dy = \int 2\,dx$

or, $y = 2x + c$

$* \quad \frac{dy}{dx} = 2x$

or, $y = x^2 + c$

$\frac{dy}{dx} = 2y$

or, $\frac{dy}{y} = 2\,dx$

or, $\int \frac{dy}{y} = \int 2\,dx$

$= \ln y = 2x + c$

$y = e^{2x} + c$

eqn (ii) ⇒

$$i\hbar \frac{d\phi}{dt} = E\phi$$

or, $i\hbar \frac{d\phi}{\phi} = E\,dt$

or, $\int \frac{d\phi}{\phi} = \int \frac{E}{i\hbar}\,dt$

or, $\ln\phi = \frac{E}{i\hbar} t + c^{\to 0}$

or, $\phi = e^{\frac{Et}{i\hbar}}$

$\phi(t) = e^{-\frac{i}{\hbar}Et}$ ⟶ (ii) এর soln এরকম পেলাম

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Genaral Solution is,

$$\boxed{\Psi(x,t) = \psi(x)\, e^{-\frac{i}{\hbar}Et}}$$

It is called Stationary state.

(i) The solution $\Psi(x,t) = \psi(x)\, e^{-\frac{i}{\hbar}Et}$ is stationary sate –

Proof:

Probability $|\Psi(x,t)|^2 =$ ~~$\psi\psi^*\psi\psi^*$~~

$= \Psi\Psi^*$

$= \psi(x)\, e^{-\frac{i}{\hbar}Et}\left(\psi^* e^{\frac{i}{\hbar}Et}\right)$

Stionary state –
(Probability is time independent

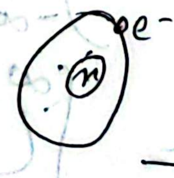
$= \psi\psi e^{0}$

$= |\psi(x)|^2 \rightarrow$ Probability does not depends on time [that's why it's called stationary state.]

(ii) They area definite total energy state —

Proof:
Total energy of operator is called Hamiltonian operator,

$\Psi_1(x,t) \rightarrow E_1$



$\Psi_1 \rightarrow E_1 = -13.6eV$

$\hat{H} = T + V$

$= \frac{1}{2}mv^2 + V$ → potential
(v → velocity)

x, x, x, $a = 0$

$= \frac{(mv)^2}{2m} + V$

Total probability = 1

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From eqn (3) we can write,

$$\boxed{\hat{H}\psi = E\psi}$$ → Eigen value eqn

$\hat{H}$ ↓ eigen state; E → eigen value

*[Hamiltonian Operation energy মাপে দেয়]

Expectation ~~of~~ Value of H,

$$\langle H\rangle = \int_{-\infty}^{\infty} H|\psi|^2 dx$$

$$= \int_{-\infty}^{\infty} (\psi^* H\psi)\, dx$$

$$= \int_{-\infty}^{\infty} (\psi^* E\psi) dx$$

$$= \int_{-\infty}^{\infty} (\psi^*\psi E)\, dx$$

$$\langle H\rangle = E\int_{-\infty}^{\infty} \psi^*\psi dx$$

$$= E$$ [as $\psi^*\psi = 1$] → Total probability = 1

$$\hat{H} = \frac{p^2}{2m} + V$$

$$\hat{H} = \left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 \cdot \frac{1}{2m} + V$$

$$\boxed{\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}}$$

$-\hbar$

$$\hat{H} = \frac{p^2}{2m} + V$$

$$\hat{H} = \left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 \frac{1}{2m} + V$$

$$\boxed{\hat{H} = \frac{-\hbar}{2m}\frac{\partial^2}{\partial x^2} + V}$$ → Hamiltonian operator

Q. Hamiltonian operator (ii)

$H\psi = \psi H$

$2x = x2$

$$\frac{d}{dx}f = f\frac{d}{dx}$$ → operator কে পিছনে লেখা যায় না।

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$$\langle H^2 \rangle = \int_{-\infty}^{\infty} (\psi^* H^2 \psi) dx$$

$$= \int_{-\infty}^{\infty} (\psi^* H \underbrace{H \psi}_{E\psi}) dx$$

$$= \int_{-\infty}^{\infty} \psi^* (HE\psi) dx$$

$$= E \int (\psi^* H \psi) dx$$

$$= E^2$$

Standard deviation,

$$\sigma = \sqrt{\langle H^2 \rangle - \langle H \rangle^2}$$

$$= \sqrt{E^2 - E^2}$$

$$= 0$$